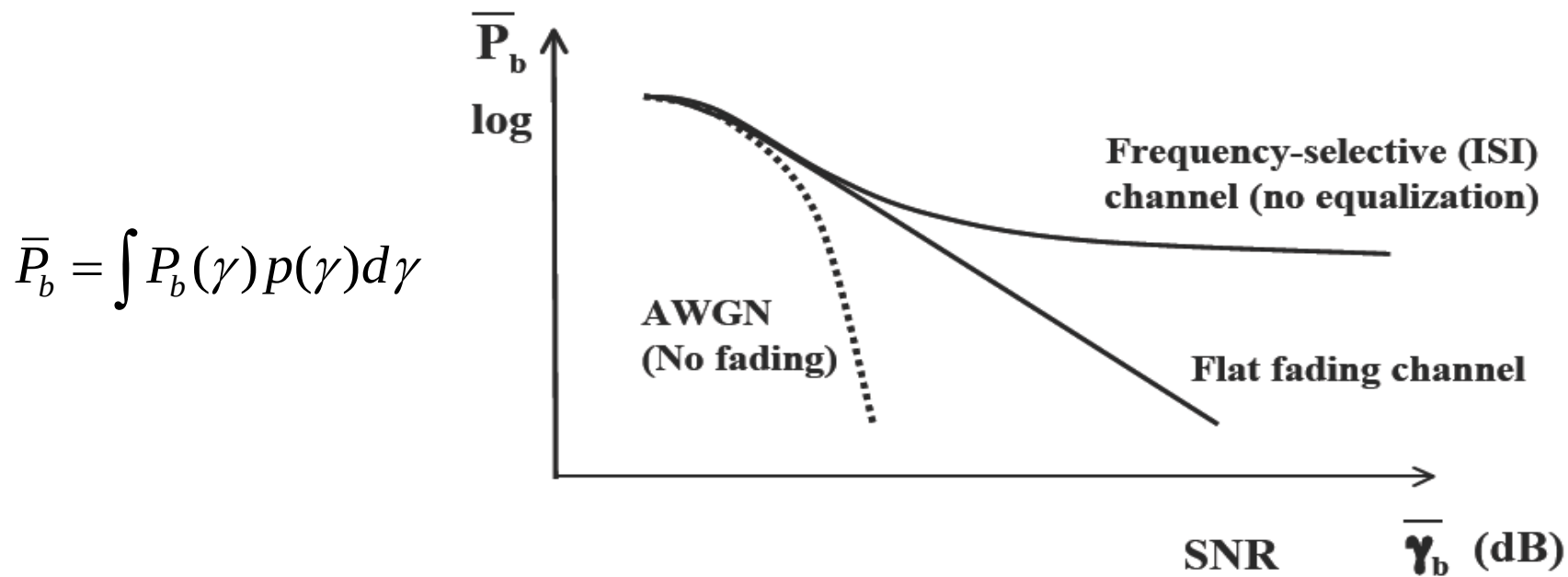
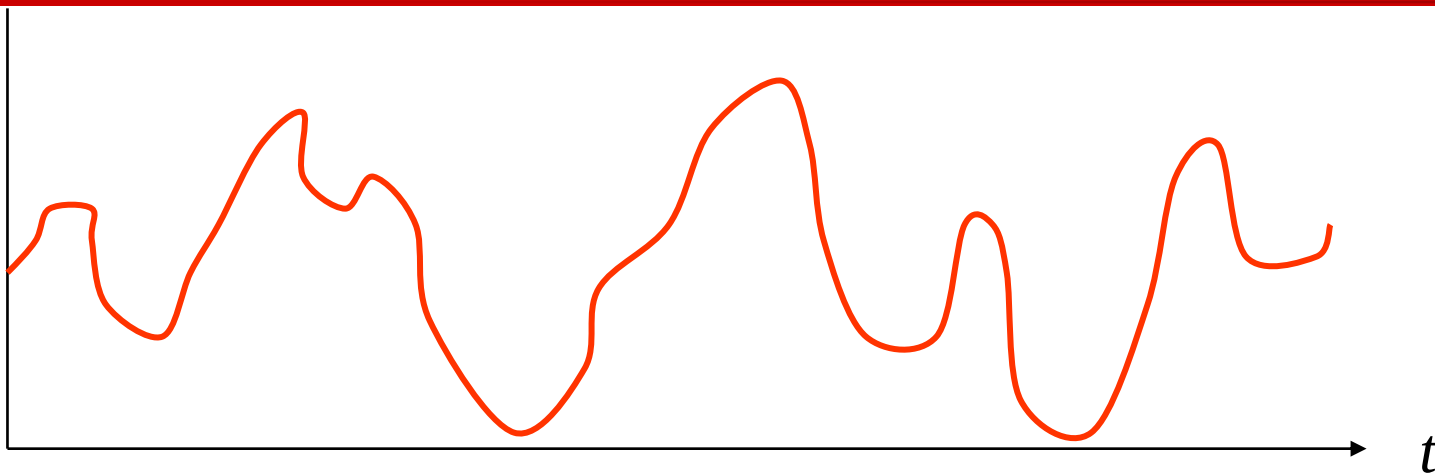

Wireless Digital Communications

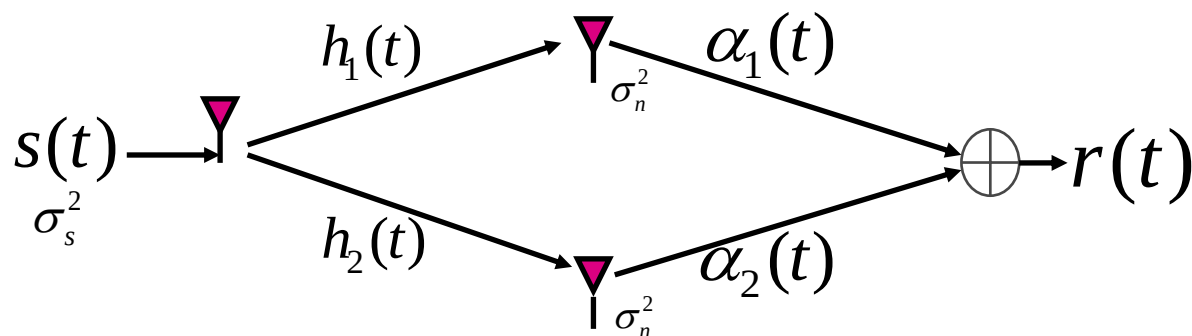
Topic 4

Diversity

基本思路



简单例子（单发两收）



$$r(t) = [\alpha_1(t)h_1(t) + \alpha_2(t)h_2(t)]s(t) + n_\Sigma(t)$$

● 输出信噪比：

$$\gamma_\Sigma = \text{SNR}(t) = \frac{\left| \sum_{i=1}^2 \alpha_i(t) h_i(t) \right|^2 \sigma_s^2}{\sigma_n^2 \sum_{i=1}^2 |\alpha_i(t)|^2} \leq \frac{\sum_{i=1}^2 |\alpha_i(t)|^2 \sum_{i=1}^2 |h_i(t)|^2}{\sum_{i=1}^2 |\alpha_i(t)|^2} \frac{\sigma_s^2}{\sigma_n^2} = \frac{\sigma_s^2}{\sigma_n^2} \sum_{i=1}^2 |h_i(t)|^2 = \sum_{i=1}^2 \gamma_i$$

● 最大信噪比条件： $\alpha_i(t) = K \times h_i^*(t)$

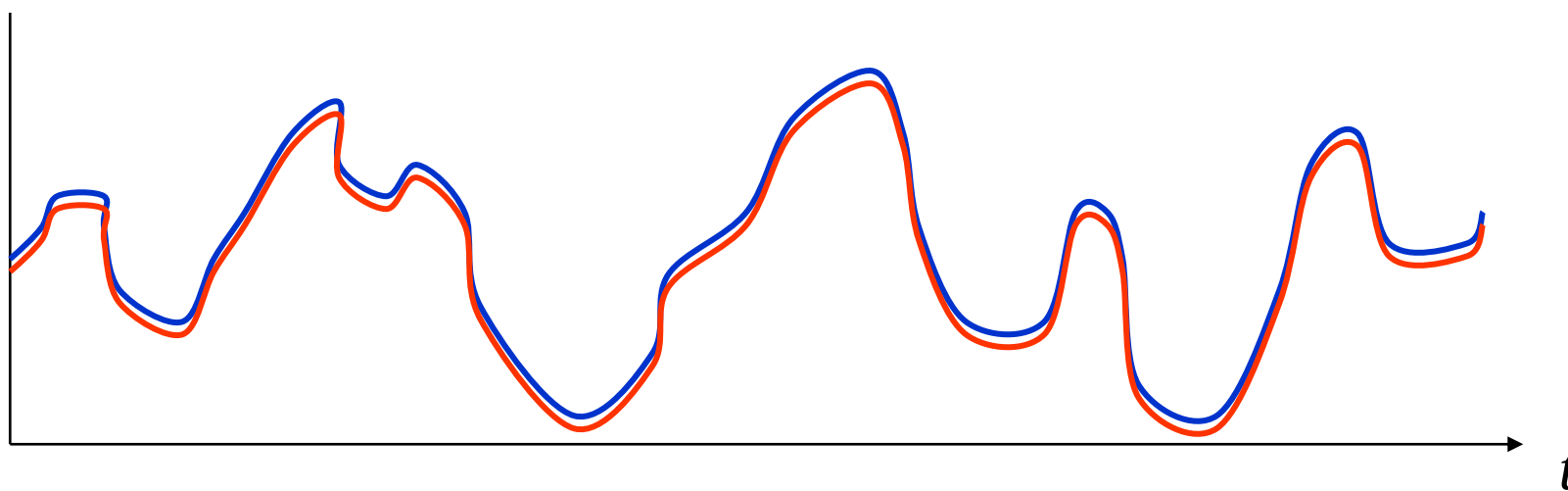
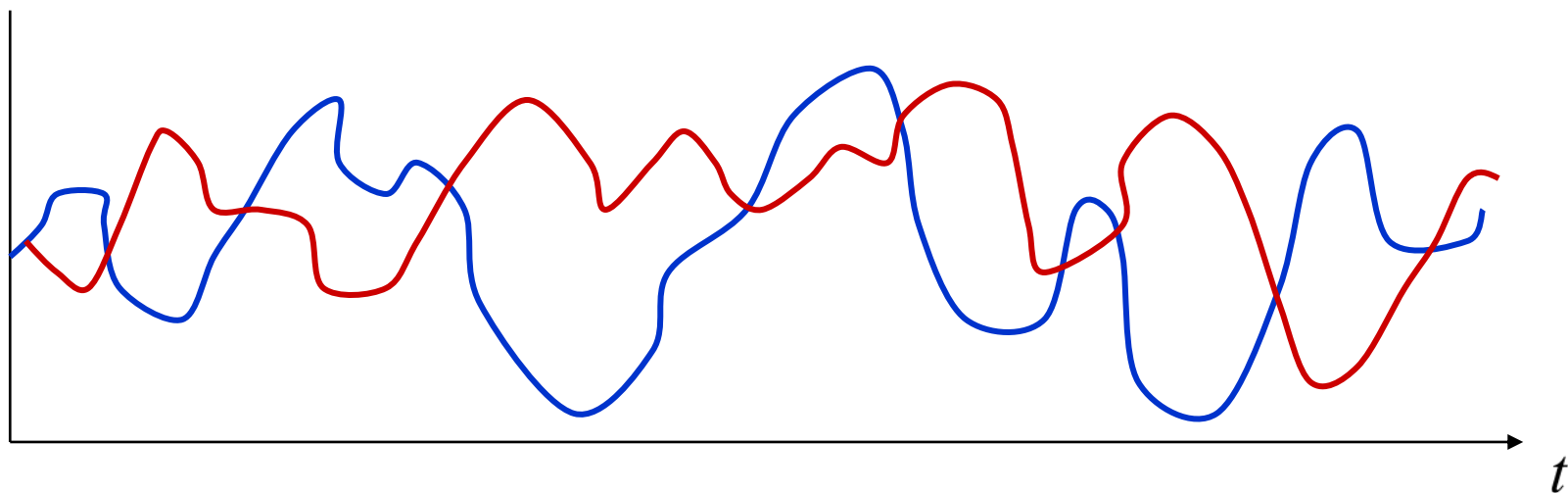
基本概念

- 阵列增益（关注平均信噪比，与天线是否独立无关）：平均信噪比增益

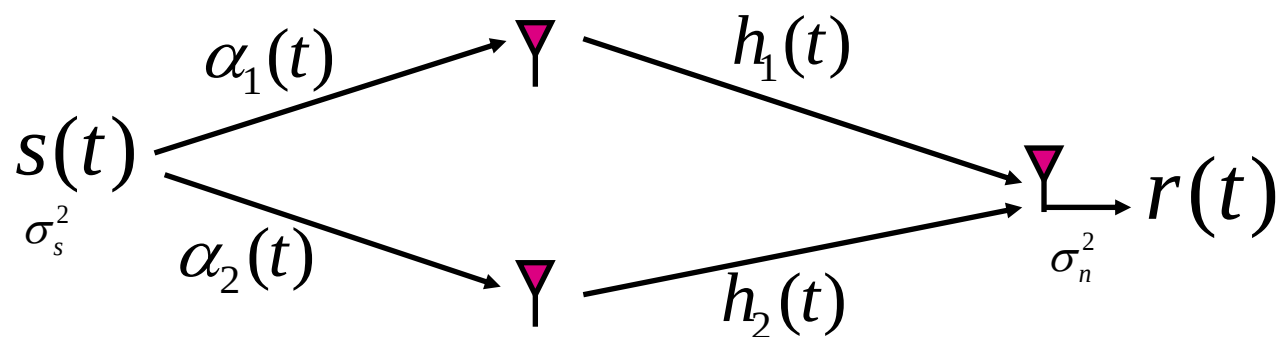
$$\overline{\gamma_{\Sigma}} = \frac{\sigma_s^2}{\sigma_n^2} \sum_{i=1}^2 E[|h_i(t)|^2] = \sum_{i=1}^2 \overline{\gamma_i} = 2 \times \overline{\gamma}$$

- 分集增益（关注瞬时信噪比，基于天线独立）：改变了合成信噪比的分布。如果天线相关，则合成信噪比波动大；如果天线独立，则合成信噪比波动小。

基本概念



简单例子（两发单收）



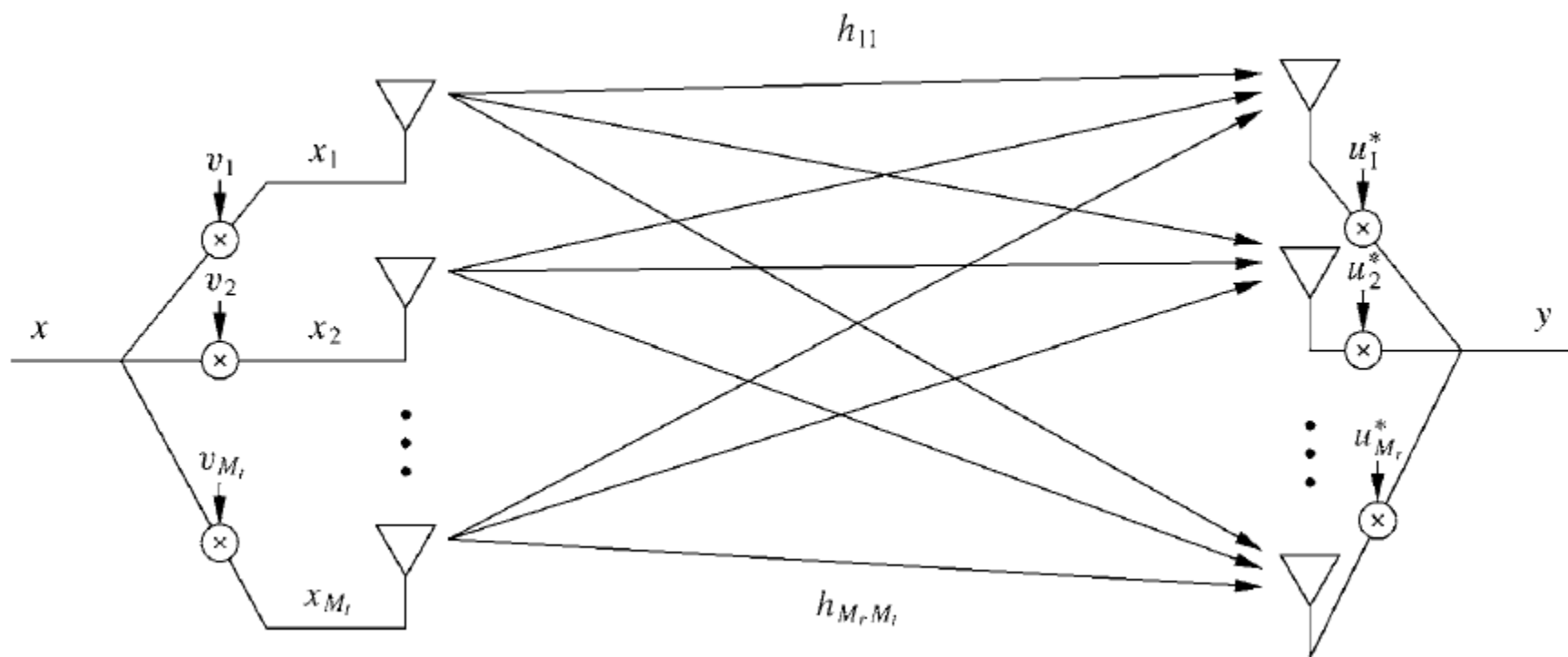
- **发射功率约束：** $|\alpha_1(t)|^2 \sigma_s^2 + |\alpha_2(t)|^2 \sigma_s^2, \quad |\alpha_1(t)|^2 + |\alpha_2(t)|^2 = 1$

$$r(t) = [\alpha_1(t)h_1(t) + \alpha_2(t)h_2(t)]s(t) + n(t)$$

$$SNR(t) = \frac{\left| \sum_{i=1}^2 \alpha_i(t)h_i(t) \right|^2 \sigma_s^2}{\sigma_n^2} \leq \sum_{i=1}^2 |\alpha_i(t)|^2 \sum_{i=1}^2 |h_i(t)|^2 \frac{\sigma_s^2}{\sigma_n^2} = \frac{\sigma_s^2}{\sigma_n^2} \sum_{i=1}^2 |h_i(t)|^2 = \sum_{i=1}^2 \gamma_i$$

- **最大信噪比条件：**
$$\begin{cases} \alpha_i(t) = K \times h_i^*(t) \\ \sum_{i=1}^2 |\alpha_i(t)|^2 = 1 \end{cases} \Rightarrow \alpha_i(t) = \frac{h_i^*(t)}{\sqrt{\sum_{i=1}^2 |h_i(t)|^2}}$$

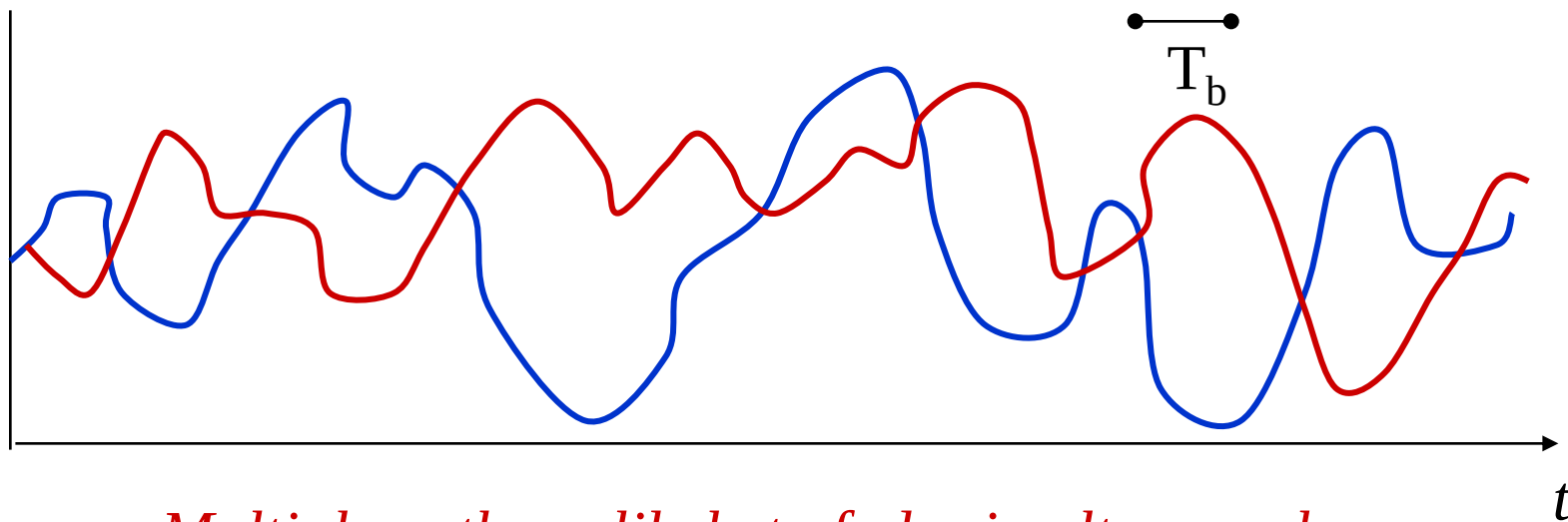
一般情况（多发多收）



Introduction to Diversity

- Basic Idea

- Send same bits over independent fading paths
 - Independent fading paths obtained by time, space, frequency, or polarization diversity
- Combine paths to mitigate fading effects



Multiple paths unlikely to fade simultaneously

Methods to Obtain Diversity Branches

- Space
 - Multiple antenna elements spaced apart by decorrelation distance. Theoretical decorrelation $\lambda/2$. Most common form of diversity. No additional power or bandwidth.
- Frequency
 - Multiple narrowband channels separated by channel coherence bandwidth. Less often used. Wasteful of scarce spectrum
- Polarization
 - Two antennas (one horizontally, the other vertically-polarized) are used. Orthogonal polarization in wireless channels exhibit uncorrelated fading. Only two-branch diversity possible. Not common.

Methods to Obtain Diversity Branches

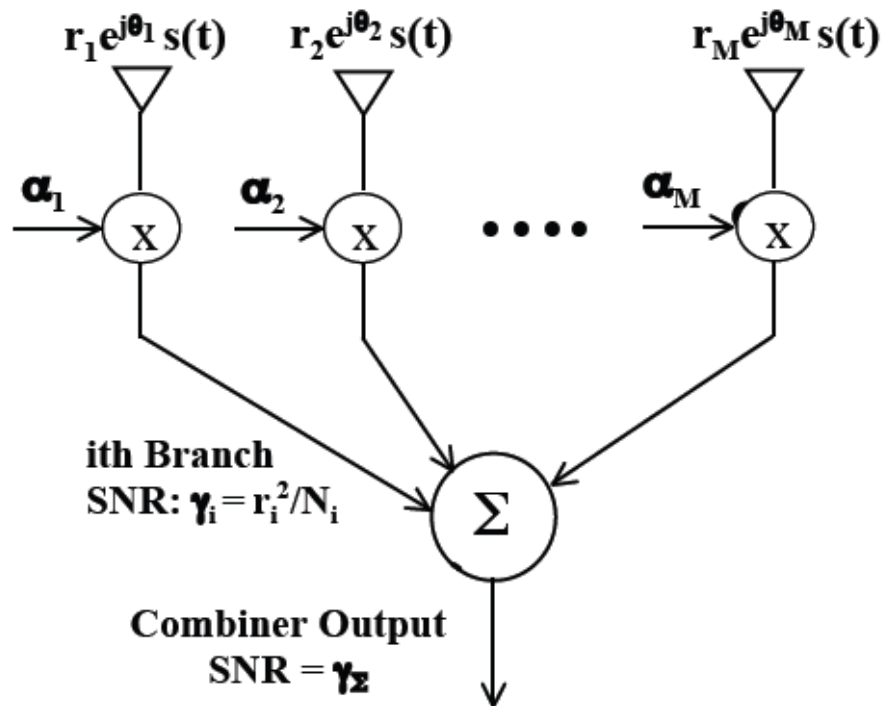
- **Angle-of-Arrival**
 - Directional antennas facing widely different directions. Scattered signal from different directions having approximately independent fading
- **Time**
 - Transmission of the same information in time slots separated by channel coherence time. Inefficient for high-speed transmissions. Useless for stationary users
- **Multipath**
 - Same as Time-diversity, except that branches are provided by channel through multipath. Takes advantage of channel provided usually undesirable multipath echoes. Principle of Rake Receivers

Combining Techniques

- Selection Combining
 - Fading path with highest gain used
- Maximal Ratio Combining
 - All paths cophased and summed with optimal weighting to maximize combiner output SNR
- Equal Gain Combining
 - All paths cophased and summed with equal weighting

System Model of Receiver Diversity

General Model



Individual branches are weighed by α_i and summed

Selection and Threshold:
all $\alpha_i = 0$ except one. Co-phasing not required

Maximal Ratio: α_i function of γ_i . Co-phasing required

Equal Gain: $|\alpha_i| = 1$.
Co-phasing required

System Model of Receiver Diversity

γ_{Σ} is a random variable with PDF $p_{\gamma_{\Sigma}}(\gamma)$ and CDF $P_{\gamma_{\Sigma}}(\gamma)$ which depends on type of fading and choice of combining

Most often PDF is obtained by differentiating CDF

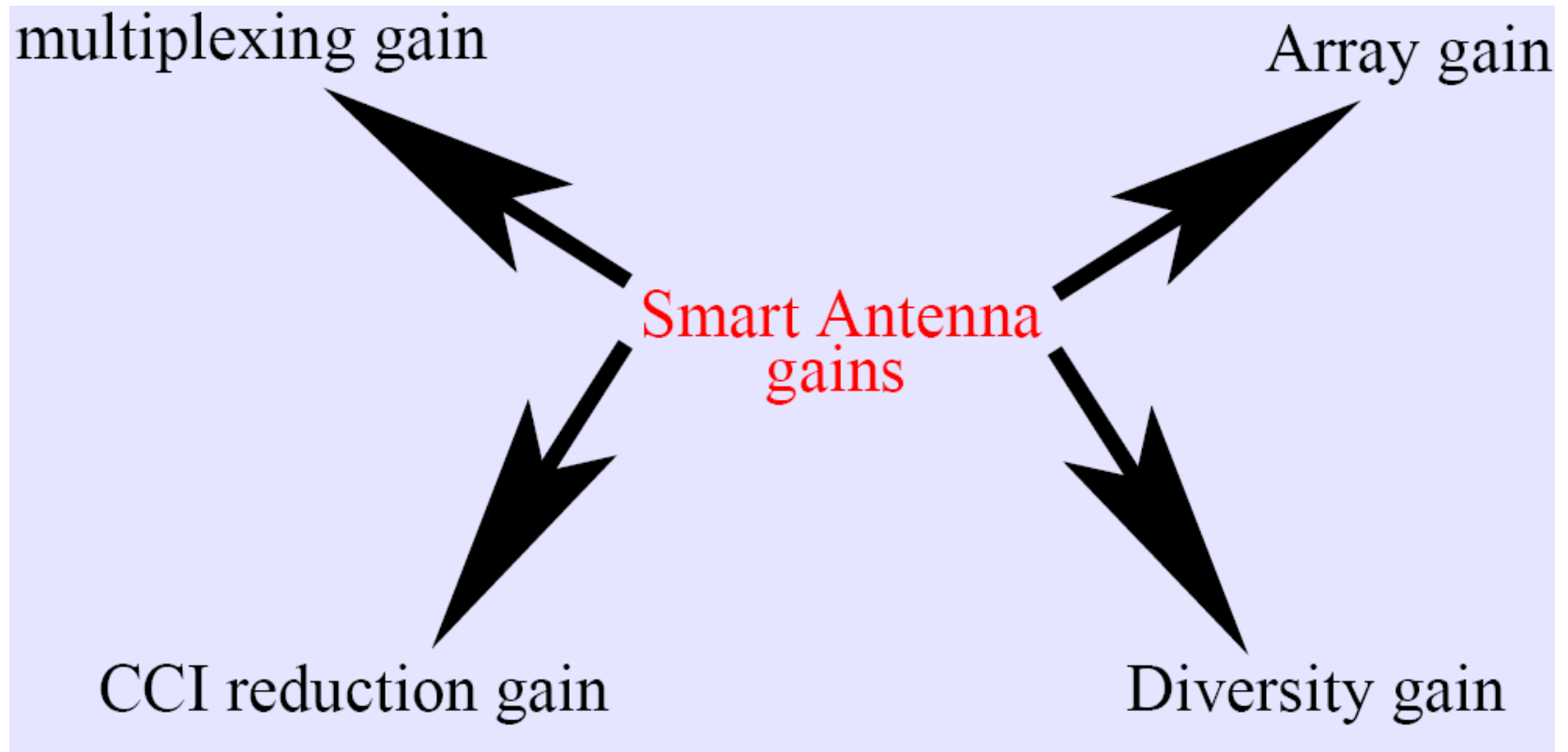
$$P_{out} = pr(\gamma_{\Sigma} < \gamma_0) = \int_0^{\gamma_0} p_{\gamma_{\Sigma}}(\gamma) d\gamma$$

$$\bar{P}_S = \int_0^{\infty} P_S(\gamma) p_{\gamma_{\Sigma}}(\gamma) d\gamma$$

$P_S(\gamma)$ is random probability of error for AWGN non-fading channel

Most often closed form solution for CDF, P_{out} and $\bar{P}_S$ unavailable. Results based on computer simulation

Gain from Using Multiple Antennas



Array and Diversity Gains

- **Array Gain**
 - Gain in SNR from coherent addition of signals and non-coherent addition (averaging) of noise over multiple antennas.
 - Gain in both fading and non-fading channels
- **Diversity Gain**
 - Gain in SNR due to elimination of weak signals (deep fades). Changes slope of probability of error.
 - Gains in fading channels

The Array Gain Principle

- The array gain is defined by the gain in mean SNR

$$S\bar{N}R = \frac{(E|h_1|^2 + \dots + E|h_N|^2)\sigma_s^2}{\sigma_n^2} = N \frac{E|h|^2\sigma_s^2}{\sigma_n^2} = N S\bar{N}R_{input}$$

$\Rightarrow$ The output SNR is N times the input SNR.

In dB, the array gain is $10 \log_{10}(N)$ dB.

The Diversity Gain Principle

- In the presence of multipath, $|h(t)|$ can follow a Rayleigh distribution (i.e. $h(t) = h_r(t) + jh_i(t)$ is complex random Gaussian)
- $|h(t)|$ fades with non-negligible probability.

$$SNR(t) = \frac{(|h_1(t)|^2 + \dots + |h_N(t)|^2)\sigma_s^2}{\sigma_n^2} \text{ is zero iff } h_i(t) \text{ is zero for all } i\text{-s!} \quad (2)$$

⇒ Probability that $SNR(t) < \epsilon$, where ϵ is a chosen small threshold decreases exponentially with N !

Some Definitions

- Diversity order used as a measure of diversity, defined as

$$M = \lim_{\bar{\gamma} \rightarrow \infty} \frac{-\log P_e}{\log \bar{\gamma}}, \quad P_e = \bar{P}_s \text{ or } P_{out}$$

- Can also use array gain (or SNR gain) $\bar{\gamma}_\Sigma / \bar{\gamma}$, where $\bar{\gamma}_\Sigma$ is the average SNR after diversity combining

Note

Specifying diversity order M and array gain A_g is roughly equivalent to saying

$$P_e \doteq (A_g \bar{\gamma})^{-M} \quad \text{for large SNR}$$

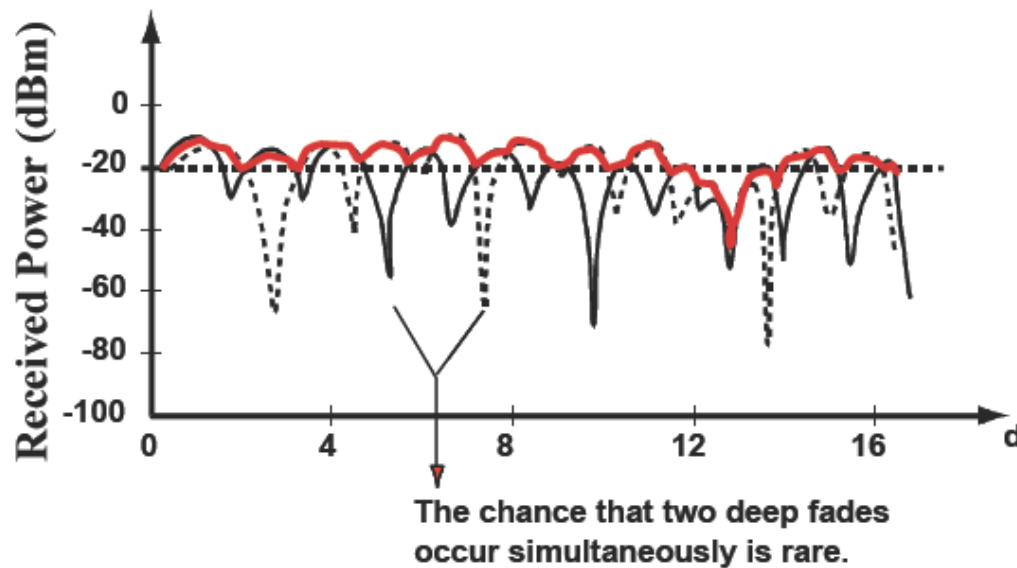
$$\log P_e = -M \log A_g - M \log \bar{\gamma}$$

Selection Combining

- Selects the path with the highest gain
- Combiner SNR is the maximum of the branch SNRs.
- CDF easy to obtain, pdf found by differentiating.
- Diminishing returns with number of antennas.
- Can get up to about 20 dB of gain.

Selection Combining

Combiner outputs the signal with highest SNR r_i^2/N_i



Since at each instant only one signal is used co-phasing is not required.

Selection Combining

Idea

Given M i.i.d. r.v., $\gamma_1, \dots, \gamma_M \geq 0$,

$$P(\max_i(\gamma_i) < c) = P(\gamma_i < c)^M$$

Outage probability

$$P_{out} = \left(1 - e^{-\frac{\gamma_0}{\bar{\gamma}}}\right)^M$$

- In Rayleigh fading $\bar{\gamma}_{\Sigma} = \bar{\gamma}(\sum_{i=1}^M 1/i)$ ($\bar{\gamma}$: average SNR at a branch)

diversity gain: M , **array gain:** $\sum_{i=1}^M 1/i$

Details

For M branch diversity, the CDF of γ_Σ is given by

$$P_{\gamma_\Sigma}(\gamma) = p(\gamma_\Sigma < \gamma) = p(\max[\gamma_1, \gamma_2, \dots, \gamma_M] < \gamma) = \prod_{i=1}^M p(\gamma_i < \gamma). \quad (7.4)$$

We obtain the pdf of γ_Σ by differentiating $P_{\gamma_\Sigma}(\gamma)$ relative to γ , and the outage probability by evaluating $P_{\gamma_\Sigma}(\gamma)$ at $\gamma = \gamma_0$. Assume that we have M branches with uncorrelated Rayleigh fading amplitudes r_i . The instantaneous SNR on the i th branch is therefore given by $\gamma_i = r_i^2/N$. Defining the average SNR on the i th branch as $\bar{\gamma}_i = E[\gamma_i]$, the SNR distribution will be exponential:

$$p(\gamma_i) = \frac{1}{\bar{\gamma}_i} e^{-\gamma_i/\bar{\gamma}_i}. \quad (7.5)$$

From (6.47), the outage probability for a target γ_0 on the i th branch in Rayleigh fading is

$$P_{out}(\gamma_0) = 1 - e^{-\gamma_0/\bar{\gamma}_i}. \quad (7.6)$$

The outage probability of the selection-combiner for the target γ_0 is then

$$P_{out}(\gamma_0) = \prod_{i=1}^M p(\gamma_i < \gamma_0) = \prod_{i=1}^M \left[1 - e^{-\gamma_0/\bar{\gamma}_i} \right]. \quad (7.7)$$

Details

If the average SNR for all of the branches are the same ($\bar{\gamma}_i = \bar{\gamma}$ for all i), then this reduces to

$$P_{out}(\gamma_0) = p(\gamma_\Sigma < \gamma_0) = \left[1 - e^{-\gamma_0/\bar{\gamma}}\right]^M. \quad (7.8)$$

Differentiating (7.8) relative to γ_0 yields the pdf for γ_Σ :

$$p_{\gamma_\Sigma}(\gamma) = \frac{M}{\bar{\gamma}} \left[1 - e^{-\gamma/\bar{\gamma}}\right]^{M-1} e^{-\gamma/\bar{\gamma}}. \quad (7.9)$$

From (7.9) we see that the average SNR of the combiner output in i.i.d. Rayleigh fading is

$$\begin{aligned} \bar{\gamma}_\Sigma &= \int_0^\infty \gamma p_{\gamma_\Sigma}(\gamma) d\gamma \\ &= \int_0^\infty \frac{\gamma M}{\bar{\gamma}} \left[1 - e^{-\gamma/\bar{\gamma}}\right]^{M-1} e^{-\gamma/\bar{\gamma}} d\gamma \\ &= \bar{\gamma} \sum_{i=1}^M \frac{1}{i}. \end{aligned} \quad (7.10)$$

BER Performance of SC

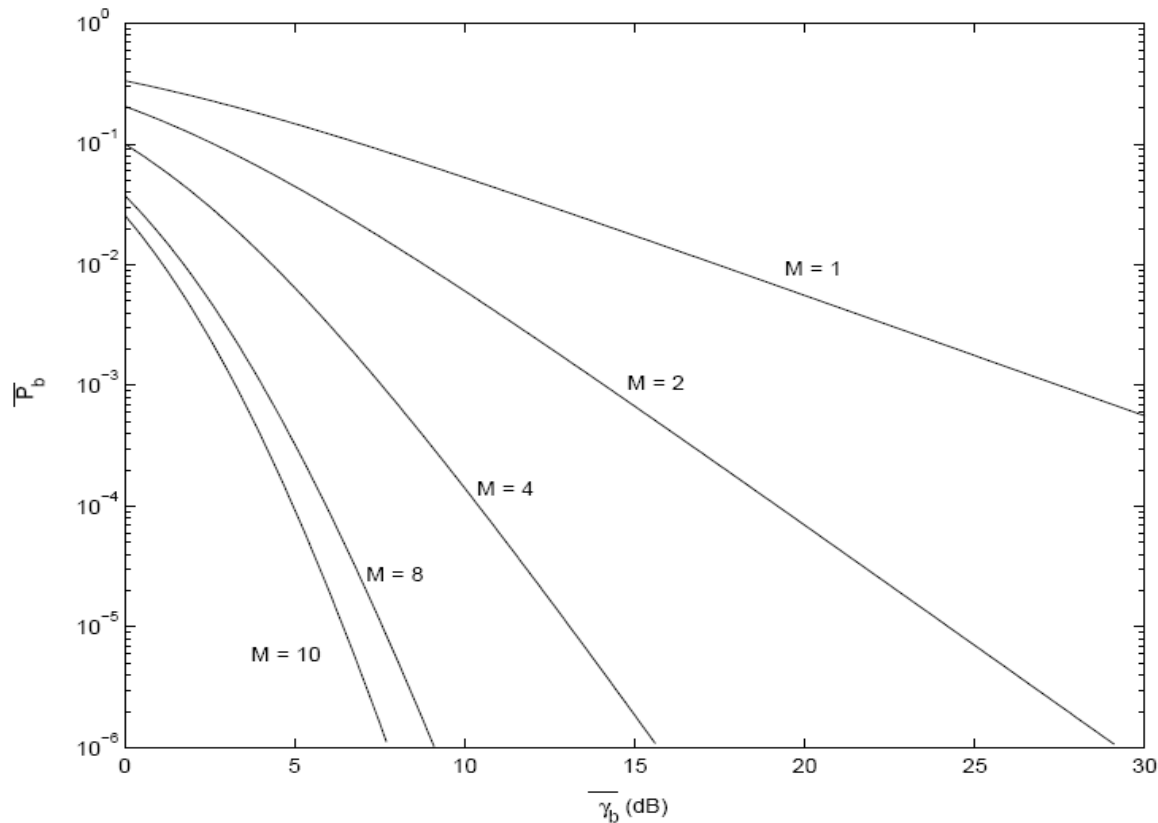


Figure 7.3: $\bar{P}_b$ of BPSK under SC with i.i.d. Rayleigh Fading.

Example

Example 7.1: Find the outage probability of BPSK modulation at $P_b = 10^{-3}$ for a Rayleigh fading channel with SC diversity for $M = 1$ (no diversity), $M = 2$, and $M = 3$. Assume equal branch SNRs of $\bar{\gamma} = 15$ dB.

Solution: A BPSK modulated signal with $\gamma_b = 7$ dB has $P_b = 10^{-3}$. Thus, we have $\gamma_0 = 7$ dB. Substituting $\gamma_0 = 10^{.7}$ and $\bar{\gamma} = 10^{1.5}$ into (7.8) yields $P_{out} = .1466$ for $M = 1$, $P_{out} = .0215$ for $M = 2$, and $P_{out} = .0031$ for $M = 3$. We see that each additional branch reduces outage probability by almost an order of magnitude.

Maximal Ratio Combining

In the general model set $\alpha_i = a_i e^{-j\theta_i}$

$$r(t) = \left[\sum_{i=1}^M \alpha_i r_i e^{j\theta_i} \right] s(t) = \left[\sum_{i=1}^M a_i r_i \right] s(t) = r \bullet s(t)$$

Assuming same noise power
at all branches:

$$\begin{aligned} \gamma_{\Sigma} &= \frac{r^2}{N_{tot}} = \frac{1}{N_0} \frac{\left(\sum_{i=1}^M a_i r_i \right)^2}{\sum_{i=1}^M a_i^2} \\ &\leq \frac{1}{N_0} \sum_{i=1}^M r_i^2 = \sum_{i=1}^M \gamma_i \end{aligned}$$

Maximizing γ_{Σ} by Cauchy-Schwartz inequality:

$$a_i^2 = r_i^2 / N_0$$

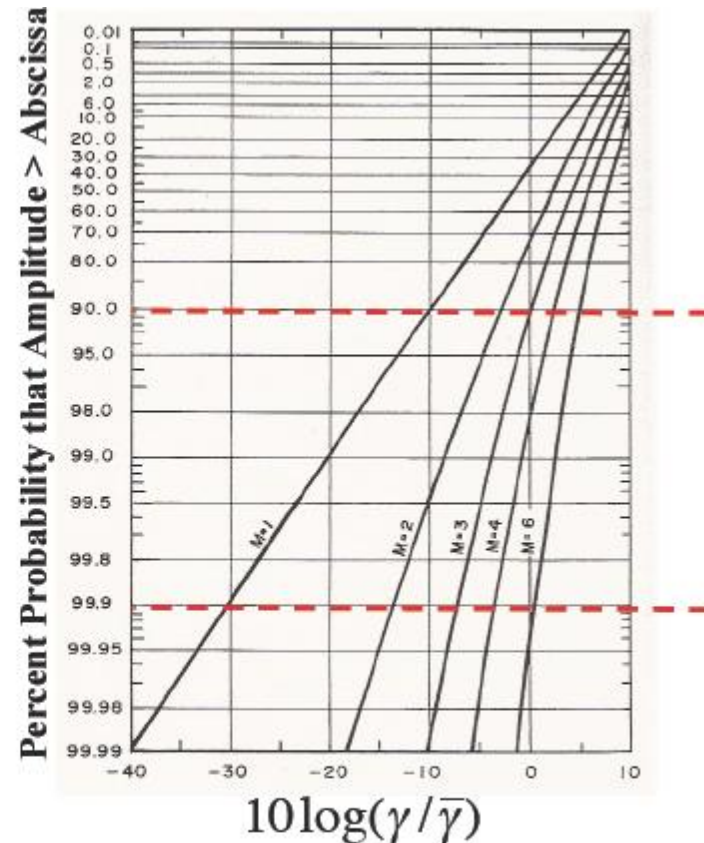
Maximal Ratio Combining

Assuming IID Rayleigh fading in each branch with equal average branch SNR $\bar{\gamma}$, resulting γ_{Σ} has chi-squared distribution with $2M$ degrees of freedom

$$\gamma_{\Sigma} = \frac{1}{N_0} \sum_{i=1}^M r_i^2 = \sum_{i=1}^M \gamma_i$$

$$p_{\gamma_{\Sigma}}(\gamma) = \frac{\gamma^{M-1} e^{-\gamma/\bar{\gamma}}}{\bar{\gamma}^M (M-1)!}, \quad \gamma \geq 0$$

$$\bar{\gamma}_{\Sigma} = M\bar{\gamma}$$



Probability distribution of SNR γ for M-branch Maximal Ratio Combining

Maximal Ratio Combining

Optimum diversity combining technique to maximize Output SNR.

$$\gamma_{\Sigma} = \sum_{i=1}^M \gamma_i$$

$$p_{\gamma_{\Sigma}}(\gamma) = \frac{\gamma^{M-1} e^{-\gamma/\bar{\gamma}}}{\bar{\gamma}^M (M-1)!}, \quad \gamma \geq 0$$

Probability of error for Modulation with SNR γ_s is:

$$P_s(\gamma_s) \approx \alpha_M Q\left(\sqrt{\beta_M \gamma_s}\right)$$

P_s for the MRC combiner output with γ_{Σ} :

$$P_s \approx \alpha_M Q\left(\sqrt{\beta_M \gamma_{\Sigma}}\right)$$

Maximal Ratio Combining

By applying the Chernoff bound to the Q function

$$Q(x) \leq e^{-x^2/2} \Rightarrow P_s \leq \alpha_M e^{-\beta_M \gamma_\Sigma / 2} = \alpha_M e^{-\beta_M \sum_{i=1}^M \gamma_i / 2}$$

$$E[P_s] \leq \alpha_M \prod_{i=1}^M E[e^{-\beta_M \gamma_i / 2}] = \alpha_M \prod_{i=1}^M \int_0^\infty e^{-\beta_M \gamma_i / 2} \frac{1}{\bar{\gamma}_i} e^{-\gamma_i / \bar{\gamma}_i} d\gamma_i$$

$$\bar{P}_s \leq \alpha_M \prod_{i=1}^M \frac{1}{(1 + \beta_M \bar{\gamma}_i / 2)}$$

For large SNR:

$$\bar{P}_s \approx \alpha_M \left(\frac{\beta_M \bar{\gamma}}{2} \right)^{-M} = c (\bar{\gamma})^{-M}$$

Diversity order is defined as slope of average probability of error as a function of SNR. For M-branch diversity its maximum is M. MRC achieves this maximum.

BER Performance of MRC

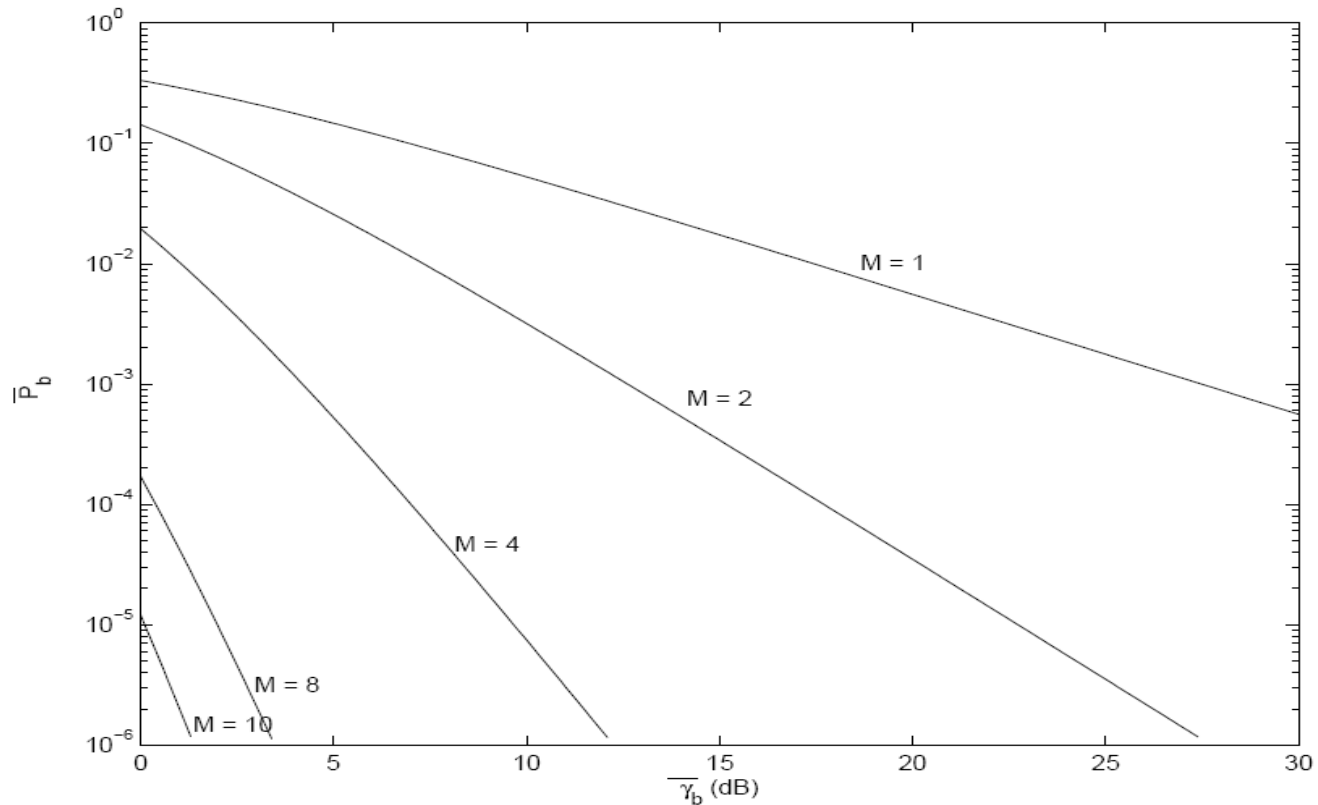


Figure 7.6: $\bar{P}_b$ for MRC with i.i.d. Rayleigh fading.

Equal Gain Combining

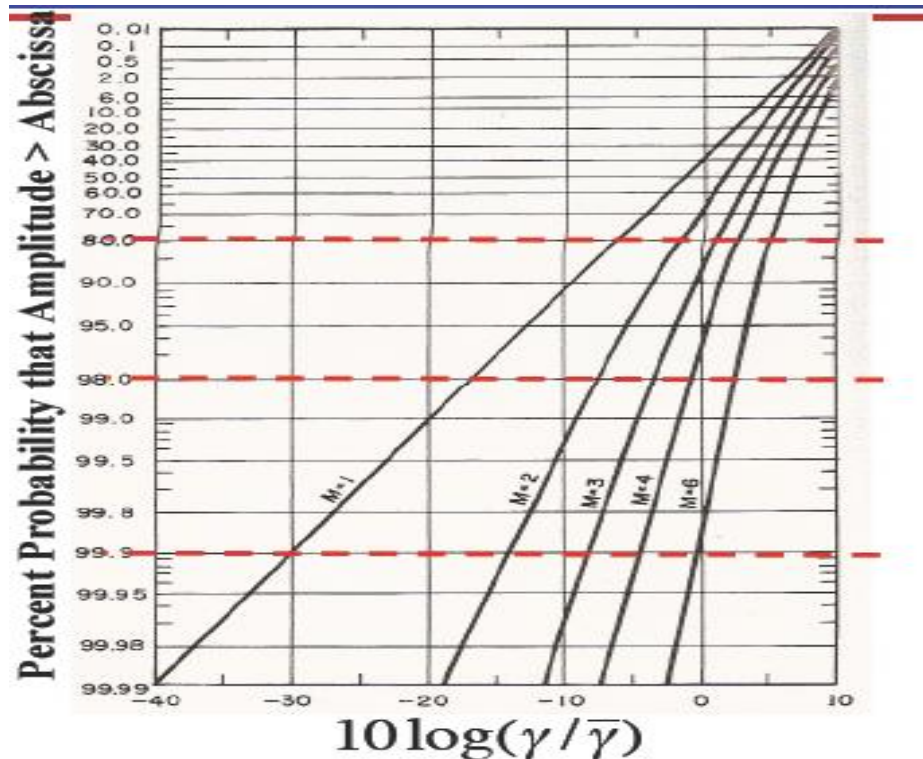
In Maximal Ratio Combining set $a_i = 1$ $\alpha_i = e^{-j\theta_i}$

$$\gamma_{\Sigma} = \frac{1}{N_0 M} \left(\sum_{i=1}^M r_i \right)^2$$

No closed-form solution for $P_{\gamma_{\Sigma}}(\gamma)$. For IID two-branch Rayleigh channel with same $\bar{\gamma}$, CDF in terms of Q function:

$$P_{\gamma_{\Sigma}}(\gamma) = 1 - e^{-2\gamma/\bar{\gamma}} - \sqrt{\pi\gamma/\bar{\gamma}} e^{-\gamma/\bar{\gamma}} (1 - 2Q(\sqrt{2\gamma/\bar{\gamma}}))$$

Equal Gain Combining

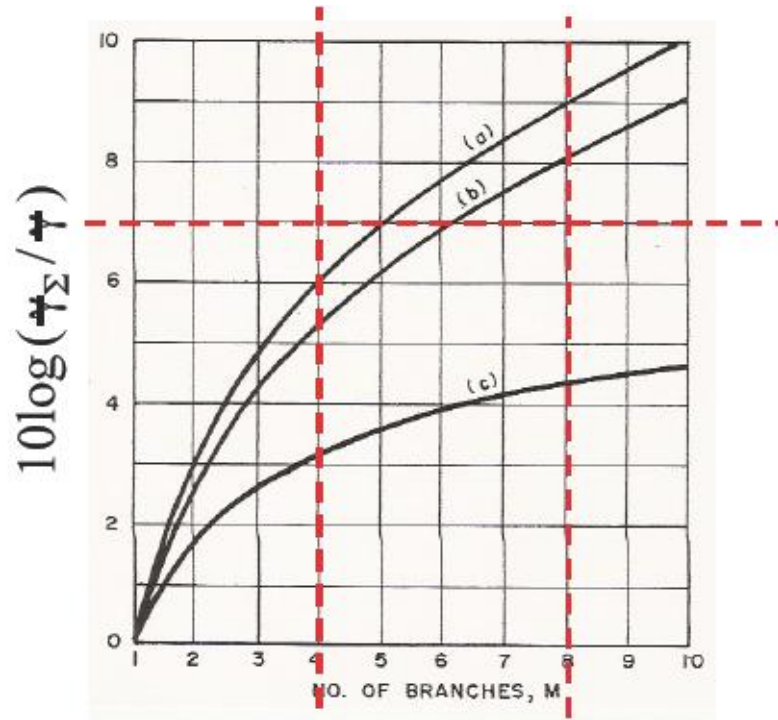


Probability distribution of SNR γ for M-branch Equal Gain Combining

SNR Improvements versus M

**For Selection Combining
SNR saturates with modest
increase in M. Maximal Ratio
and Equal Gain do not exhibit
Saturation, but slope changes**

**Maximal Ratio and Equal Gain
outperform Selection; however,
their relative performance is close
(within 1 dB). Equal Gain simpler
to implement**



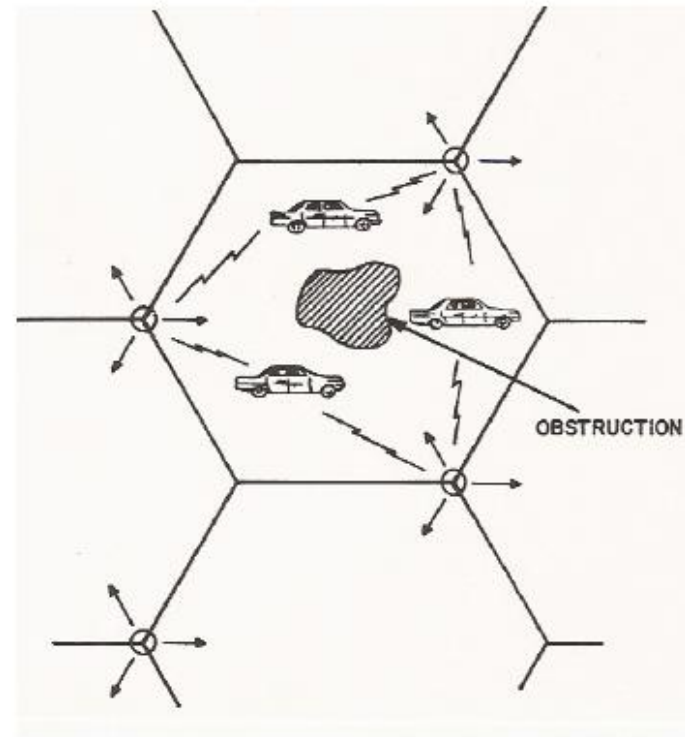
**Improvements in average SNR for M-Branch
compared to one branch. (a) Maximal Ratio,
(b) Equal Gain, (c) Selection**

Macroscopic Diversity

A technique to reduce slow fading (shadowing) effects

Inherent in cellular architecture. Signal can be received from more than one base station. A form of Selection Combining

Fading is lognormal. Greater improvement for larger standard deviations



Transmit Diversity

- Multiple transmit antennas, with power divided among them
- Suitable for systems with greater capabilities at the transmit site (example: cellular)
- Implementation depends on channel knowledge
 - With transmitter channel knowledge (CSIT), performance is similar to receiver diversity (same array/diversity gain)
 - Without channel knowledge, can obtain diversity gain (but not array gain) through Alamouti scheme: transmission over space and time (2 consecutive symbols)

Channel Known at Transmitter

We assume path gain $r_i e^{j\theta_i}$ is known at transmitter

$s(t)$ transmitted signal with energy per symbol E_s is
Multiplied by constant complex gain α_i

$$\alpha_i = a_i e^{-j\theta_i}, \quad 0 \leq a_i \leq 1$$

Total energy constraint E_s requires $\sum_{i=1}^M a_i^2 = 1$

Signals transmitted from M antennas
are combined in the air:

$$r(t) = \sum_{i=1}^M a_i r_i s(t)$$

Channel Known at Transmitter

Similar to MRC coefficients are found to maximize SNR:

$$a_i = \frac{r_i}{\sqrt{\sum_{i=1}^M r_i^2}}$$

Resulting SNR:

$$\gamma_{\Sigma} = \frac{E_s}{N_0} \sum_{i=1}^M r_i^2 = \sum_{i=1}^M \gamma_i$$

$$\bar{\gamma}_{\Sigma} = M\bar{\gamma}$$

$$P_s \approx \alpha_M Q(\sqrt{\beta_M \gamma_{\Sigma}}) \leq \alpha_M e^{-\beta_M \gamma_{\Sigma}/2} = \alpha_M \exp[-(\beta_M \sum_{i=1}^M \gamma_i)/2]$$

$$\bar{P}_s \leq \alpha_M \prod_{i=1}^M \frac{1}{(1 + \beta_M \bar{\gamma}_i/2)}$$

for large SNR:

$$\bar{P}_s \approx \alpha_M \left(\frac{\beta_M \bar{\gamma}}{2} \right)^{-M} = c(\bar{\gamma})^{-M}$$

Similar to MRC full diversity order of M is achieved

Channel Unknown at Transmitter

The Alamouti Scheme

Transmission in Space and Time

First Symbol Period

antenna 1 s_1

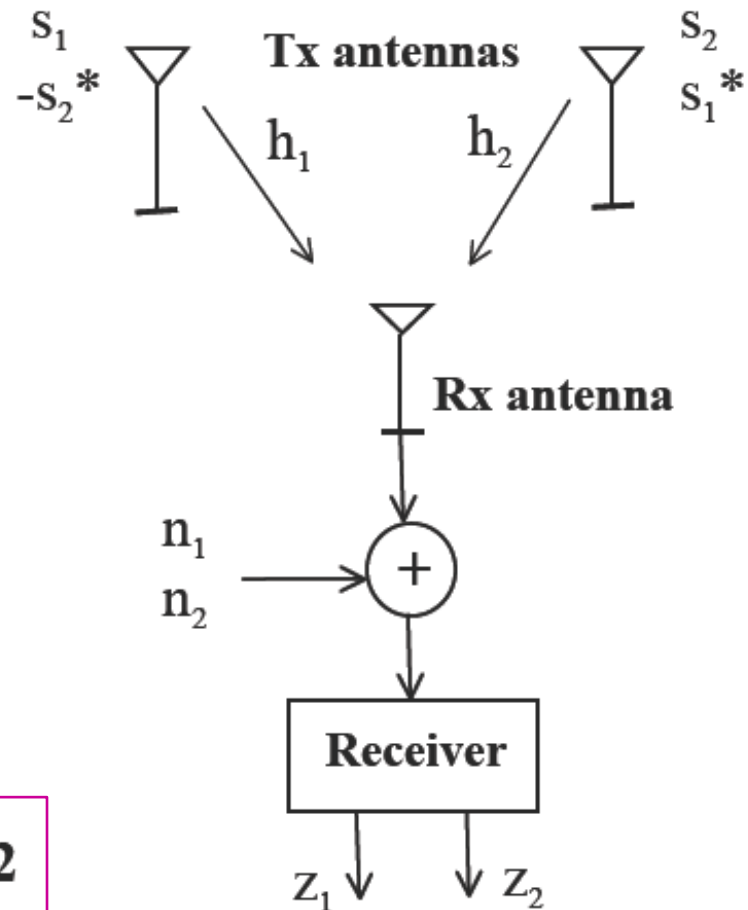
antenna 2 s_2

Second Symbol Period

antenna 1 $-s_2^*$

antenna 2 s_1^*

Each Transmission has energy $E_s/2$



The Alamouti Scheme

Channel Gains: $h_1 = r_1 e^{j\theta_2} \quad , \quad h_2 = r_2 e^{j\theta_2}$

Received Signal: $\mathbf{y} = \begin{bmatrix} y_1 & y_2^* \end{bmatrix}^T$

$$\begin{aligned} y_1 &= h_1 s_1 + h_2 s_2 + n_1 \\ y_2 &= -h_1 s_2^* + h_2 s_1^* + n_2 \end{aligned}$$

$$\mathbf{y} = \begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = \mathbf{H}_A \mathbf{s} + \mathbf{n}$$

$$\mathbf{s} = \begin{bmatrix} s_1 & s_2 \end{bmatrix}^T \quad \mathbf{n} = \begin{bmatrix} n_1 & n_2 \end{bmatrix}^T \quad \mathbf{H}_A = \begin{bmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{bmatrix}$$

The Alamouti Scheme

Define: $\mathbf{z} = \mathbf{H}_A^H \mathbf{y}$ where $\mathbf{H}_A^H$ is conjugate Transpose of $\mathbf{H}_A$

$$\mathbf{H}_A^H \mathbf{H}_A = (|h_1|^2 + |h_2|^2) \mathbf{I}_2$$

$$\mathbf{z} = \begin{bmatrix} z_1 & z_2 \end{bmatrix}^T = (|h_1|^2 + |h_2|^2) \mathbf{I}_2 \mathbf{s} + \tilde{\mathbf{n}}$$

$$\tilde{\mathbf{n}} = \mathbf{H}_A^H \mathbf{n} \quad E[\tilde{\mathbf{n}} \tilde{\mathbf{n}}^*] = (|h_1|^2 + |h_2|^2) N_0 \mathbf{I}_2$$

The Alamouti Scheme

$$z_i = \left(|h_1|^2 + |h_2|^2 \right) s_i + \tilde{n}_i, \quad i = 1, 2$$

$$\gamma_i = \frac{\left(|h_1|^2 + |h_2|^2 \right) E_s}{2N_0}$$

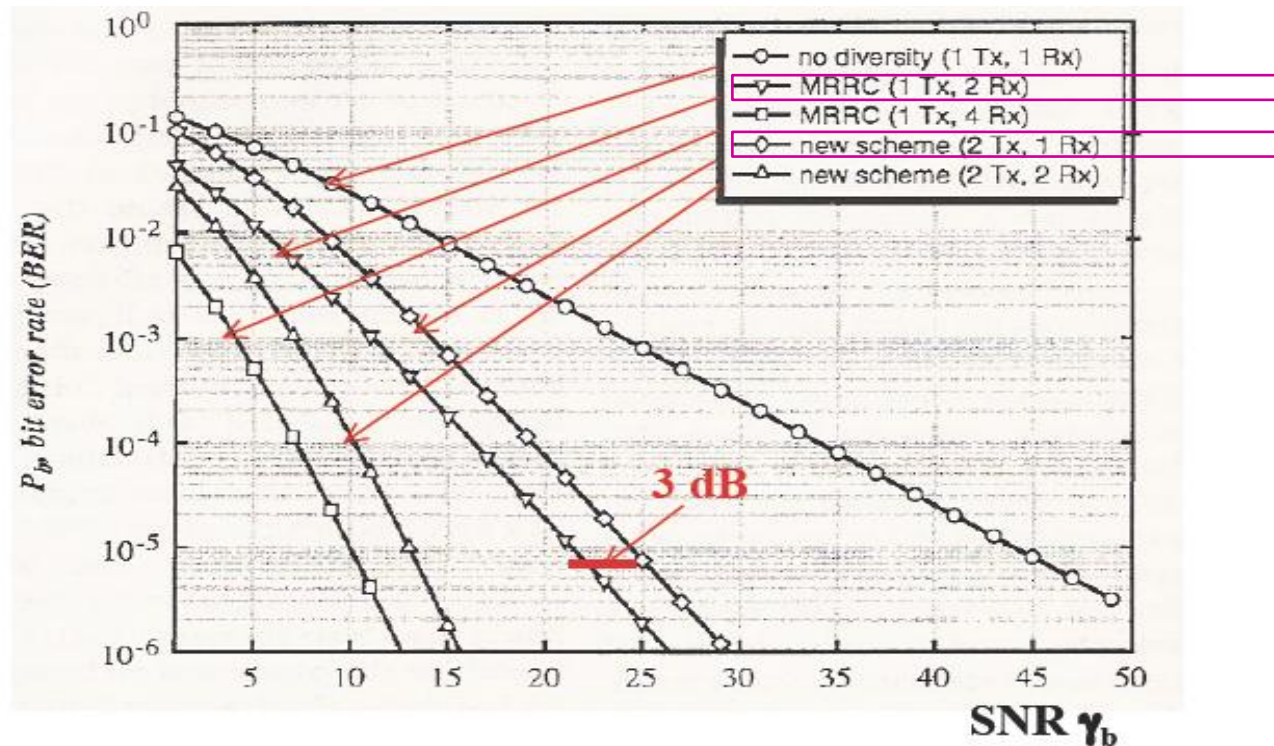
factor 2 represents transmission of half energy $E_s/2$

$$E[\gamma_i] = E[|h|^2] \bullet E_s / N_0$$

Observations:

- 1. Alamouti scheme achieves a diversity order of 2 although channel knowledge not available at transmitter**
- 2. Alamouti scheme achieves an array gain of 1**

Performance of Alamouti Scheme



BER performance of Alamouti Scheme: comparison of BPSK with Maximal Ratio and two-branch transmit diversity in Rayleigh fading

Main Points

- Diversity overcomes the effects of flat fading by combining multiple independent fading paths
- Diversity typically entails some penalty in terms of rate, bandwidth, complexity, or size
- Different combining techniques offer different levels of complexity and performance
 - ▲ MRC provides best diversity performance
 - ▲ EGC easier to implement than MRC: hard to analyze; performance about 1dB worse than MRC
- Macroscopic Diversity, inherent in cellular system architecture reduces slow fading (shadowing) effects
- Performance of Transmit Diversity depends on channel knowledge
 - ▲ With CSIT same performance as receiver diversity
 - ▲ Without CSIT Alamouti scheme provides same diversity gain, but array gain is 3dB lower

-
- Ex. 7.7, 7.9, 7.16